

《数据库系统原理》实验报告（1）																										
题目：DDL 语言实验																										
学号	2452769	姓名	幸可函	日期	2026/3/24																					
实验环境： 1. 操作系统：Windows 11 2. 容器平台：Docker Desktop for Windows（版本：4.65.0） 3. 数据库镜像：oceanbase/oceanbase-ce:latest 4. 数据库系统：OceanBase Community Edition 4.3.5.5																										
实验步骤及结果截图： 1. 参照下表内容，在数据库中建立教学管理系统相关数据表。																										
<table border="1"><thead><tr><th colspan="3">表名：Student</th></tr><tr><th>字段</th><th>类型</th><th>约束</th></tr></thead><tbody><tr><td>No</td><td>Integer</td><td>主键，非空</td></tr><tr><td>Name</td><td>Varchar</td><td>非空</td></tr><tr><td>Gender</td><td>Varchar</td><td>非空，只能为 F 或者 M</td></tr><tr><td>Age</td><td>Integer</td><td>非空</td></tr><tr><td>D_no</td><td>Integer</td><td>非空，外键</td></tr></tbody></table>						表名：Student			字段	类型	约束	No	Integer	主键，非空	Name	Varchar	非空	Gender	Varchar	非空，只能为 F 或者 M	Age	Integer	非空	D_no	Integer	非空，外键
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字段	类型	约束																								
S_no	Integer	非空，外键																								
C_no	Integer	非空，外键																								
Score	Integer	非空																								
实验结果截图： 																										

```
obclient(root@sys)[teaching_management]> create table dept(
-> No int not null,
-> Name varchar(15) not null,
-> primary key (No));
Query OK, 0 rows affected (0.116 sec)
```

```
obclient(root@sys)[teaching_management]> create table course(
-> No int not null,
-> Name varchar(15) not null,
-> Credit int not null,
-> primary key(No));
Query OK, 0 rows affected (0.099 sec)
```

```
obclient(root@sys)[teaching_management]> create table score(
-> S_no int not null,
-> C_no int not null,
-> Score int not null,
-> foreign key (S_no) references student(No),
-> foreign key (C_no) references course(No));
Query OK, 0 rows affected (0.185 sec)
```

2.使用以下语句查询表结构：DESC <表名>

```
obclient(root@sys)[teaching_management]> desc student;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| No    | int(11)       | NO   | PRI | NULL    |       |
| Name  | varchar(15)   | NO   |     | NULL    |       |
| Gender | varchar(1)    | NO   |     | NULL    |       |
| Age   | int(11)       | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.006 sec)
```

```
obclient(root@sys)[teaching_management]> desc dept;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| No    | int(11)       | NO   | PRI | NULL    |       |
| Name  | varchar(15)   | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.003 sec)
```

```
obclient(root@sys)[teaching_management]> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| No    | int(11)       | NO   | PRI | NULL    |       |
| Name  | varchar(15)   | NO   |     | NULL    |       |
| Credit | int(11)       | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.004 sec)
```

```
obclient(root@sys)[teaching_management]> desc score;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| S_no  | int(11)       | NO   |     | NULL    |       |
| C_no  | int(11)       | NO   |     | NULL    |       |
| Score | int(11)       | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.004 sec)
```

3. 使用以下语句查询建立的约束:

```
SELECT table_name, constraint_name, constraint_type
FROM information_schema.TABLE_CONSTRAINTS
WHERE table_name=<表名>
```

```
obclient(root@sys)[teaching_management]> SELECT table_name, constraint_name, constraint_type FROM information_s
hema.TABLE_CONSTRAINTS WHERE table_name='student';
+-----+-----+-----+
| table_name | constraint_name | constraint_type |
+-----+-----+-----+
| student   | PRIMARY         | PRIMARY KEY    |
| student   | student_OBCHECK_1774341836809447 | CHECK          |
+-----+-----+-----+
2 rows in set (0.146 sec)
```

```
obclient(root@sys)[teaching_management]> SELECT table_name, constraint_name, constraint_type FROM information_s
hema.TABLE_CONSTRAINTS WHERE table_name='dept';
+-----+-----+-----+
| table_name | constraint_name | constraint_type |
+-----+-----+-----+
| dept      | PRIMARY        | PRIMARY KEY    |
+-----+-----+-----+
1 row in set (0.013 sec)
```

```
obclient(root@sys)[teaching_management]> SELECT table_name, constraint_name, constraint_type FROM information_s
hema.TABLE_CONSTRAINTS WHERE table_name='course';
+-----+-----+-----+
| table_name | constraint_name | constraint_type |
+-----+-----+-----+
| course     | PRIMARY        | PRIMARY KEY    |
+-----+-----+-----+
1 row in set (0.014 sec)
```

```
obclient(root@sys)[teaching_management]> SELECT table_name, constraint_name, constraint_type FROM information_s
hema.TABLE_CONSTRAINTS WHERE table_name='score';
+-----+-----+-----+
| table_name | constraint_name | constraint_type |
+-----+-----+-----+
| score      | score_OBFK_1774342063237142 | FOREIGN KEY    |
| score      | score_OBFK_1774342063237554 | FOREIGN KEY    |
+-----+-----+-----+
2 rows in set (0.038 sec)
```

4.将 student 中的 name 由 varchar 改为 char(15)

```
obclient(root@sys)[teaching_management]> alter table student modify column Name char(15) not null;
Query OK, 0 rows affected (0.354 sec)
```

```
obclient(root@sys)[teaching_management]> desc student;
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| No    | int(11)   | NO   | PRI | NULL    |       |
| Name  | char(15)  | NO   |     | NULL    |       |
| Gender | varchar(1) | NO   |     | NULL    |       |
| Age   | int(11)   | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.003 sec)
```

5.使用 alter 添加 student 和 course 的 d_no 列及其约束。

```
obclient(root@sys)[teaching_management]> alter table student add column D_no int not null;
Query OK, 0 rows affected (0.160 sec)
```

```
obclient(root@sys)[teaching_management]> alter table student add foreign key (D_no) references dept(No);
Query OK, 0 rows affected (0.430 sec)
```

```
obclient(root@sys)[teaching_management]> desc student;
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| No    | int(11)   | NO   | PRI | NULL    |       |
| Name  | char(15)  | NO   |     | NULL    |       |
| Gender | varchar(1) | NO   |     | NULL    |       |
| Age   | int(11)   | NO   |     | NULL    |       |
| D_no  | int(11)   | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.005 sec)
```

```
obclient(root@sys)[teaching_management]> alter table course add column D_no int not null;
Query OK, 0 rows affected (0.077 sec)

obclient(root@sys)[teaching_management]> alter table course add foreign key (D_no) references dept(No);
Query OK, 0 rows affected (0.289 sec)

obclient(root@sys)[teaching_management]> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| No    | int(11)   | NO   | PRI | NULL    |       |
| Name  | varchar(15)| NO   |     | NULL    |       |
| Credit| int(11)   | NO   |     | NULL    |       |
| D_no  | int(11)   | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.003 sec)
```

6. 将 score 表中所有约束删除后再建立一次

(1) 删除外码约束

```
obclient(root@sys)[teaching_management]> alter table score drop foreign key score_OBFK_1774342063237142;
Query OK, 0 rows affected (0.082 sec)
```

Database changed

```
obclient(root@sys)[teaching_management]> alter table score drop foreign key score_OBFK_1774342063237554;
Query OK, 0 rows affected (0.106 sec)
```

结果:

```
obclient(root@sys)[teaching_management]> SELECT table_name, constraint_name, constraint_type FROM information_schema.TABLE_
CONSTRAINTS WHERE table_name='score';
Empty set (0.017 sec)
```

(2) 删除非空约束:

```
obclient(root@sys)[teaching_management]> alter table score modify column S_no int;
Query OK, 0 rows affected (0.069 sec)
```

```
obclient(root@sys)[teaching_management]> alter table score modify column C_no int;
Query OK, 0 rows affected (0.085 sec)
```

```
obclient(root@sys)[teaching_management]> alter table score modify column Score int;
Query OK, 0 rows affected (0.074 sec)
```

```
obclient(root@sys)[teaching_management]> desc score;
+-----+-----+-----+-----+-----+-----+
| Field | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| S_no  | int(11)   | YES  |     | NULL    |       |
| C_no  | int(11)   | YES  |     | NULL    |       |
| Score | int(11)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.003 sec)
```

(3) 再建立:

```
obclient(root@sys)[teaching_management]> alter table score modify column Score int not null;
Query OK, 0 rows affected (0.292 sec)

obclient(root@sys)[teaching_management]> alter table score modify column C_no int not null;
Query OK, 0 rows affected (0.302 sec)

obclient(root@sys)[teaching_management]> alter table score modify column S_no int not null;
Query OK, 0 rows affected (0.293 sec)

obclient(root@sys)[teaching_management]> desc score;
+-----+-----+-----+-----+-----+-----+
| Field | Type   | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| S_no  | int(11)| NO   |     | NULL    |       |
| C_no  | int(11)| NO   |     | NULL    |       |
| Score | int(11)| NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.004 sec)
```

```
obclient(root@sys)[teaching_management]> alter table score add foreign key (S_no) references student(No);
Query OK, 0 rows affected (0.285 sec)

obclient(root@sys)[teaching_management]> alter table score add foreign key (C_no) references course(No);
Query OK, 0 rows affected (0.401 sec)

obclient(root@sys)[teaching_management]> SELECT table_name, constraint_name, constraint_type FROM information_schema.TABLE_
CONSTRAINTS WHERE table_name='score';
+-----+-----+-----+-----+-----+-----+
| table_name | constraint_name | constraint_type |
+-----+-----+-----+-----+-----+-----+
| score      | score_OBFK_1774343319185121 | FOREIGN KEY     |
| score      | score_OBFK_1774343332614718 | FOREIGN KEY     |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.012 sec)
```

出现的问题:

(1) 建表时报语法错误 syntax error:

```
obclient(root@sys)[teaching_management]> create table student(
    -> No int not null,
    -> Name varchar not null,
    -> Gender varchar not null,
    -> Age int not null,
    -> primary key(No),
    -> check (Gender in ('F','M')));
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your OceanBase ve
rsion for the right syntax to use near 'not null,
Gender varchar not null,
Age int not null,
primary key(No),
check (Gen' at line 3
[172.17.0.2:2882] [2026-03-24 08:43:00.696136] [YB42AC110002-00064DC06E9D8C34-0-0]
```

(2) 试图删除外码约束时报错:

```
obclient(root@sys)[teaching_management]> alter table score drop foreign key S_no;
ERROR 1091 (42000): Can't DROP 'S_no'; check that column/key exists
[172.17.0.2:2882] [2026-03-24 09:02:05.515211] [YB42AC110002-00064DC06E6DCF19-0-0]
```

解决方案:

(1) 在建立数据表时, varchar 型数据表需指定最大长度 n 如 varchar(n)

```
obclient(root@sys)[teaching_management]> create table student( No int not null, Name varchar(15) not null, Gende
r varchar(1) not null, Age int not null, primary key(No), check (Gender in ('F','M')));
Query OK, 0 rows affected (0.261 sec)
```

(2) 删外码约束时要删除约束的名字 constraint_name (由(3)查得) 而不是外码本身

```
obclient(root@sys)[teaching_management]> alter table score drop foreign key score_0BFK_1774342063237142;  
Query OK, 0 rows affected (0.082 sec)
```